

Sample L^AT_EX Document

Your Name

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Abstract

This is a sample L^AT_EX document demonstrating various features including mathematical equations, figures, tables, code listings, and bibliography references. Use this as a template to get started with your own documents. Now I have added more text to the abstract to ensure it spans multiple lines and provides a better overview of the document's content and purpose. Change 3. Change 4. Change 5.

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1 Introduction

Welcome to L^AT_EX! This document serves as a starting point for your L^AT_EX journey in VS Code. L^AT_EX is a powerful typesetting system widely used in academia and scientific publishing.

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2 Mathematics

L^AT_EX excels at typesetting mathematical equations.

2.1 Inline Math

Einstein's famous equation is $E = mc^2$. The quadratic formula is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

2.2 Display Math

The Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \quad (1)$$

The Euler's identity:

$$e^{i\pi} + 1 = 0 \quad (2)$$

A system of equations:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad (3)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (4)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (5)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (6)$$

3 Lists

3.1 Itemized List

- First item
- Second item

- Third item with sub-items:
 - Sub-item A
 - Sub-item B

3.2 Enumerated List

1. Install a $\text{\LaTeX}$ distribution (MiKTeX, TeX Live, or MacTeX)
2. Install the LaTeX Workshop extension in VS Code
3. Create a `.tex` file
4. Write your document
5. Compile and preview

4 Tables

Table 1: Sample Table

Left	Center	Right
Item 1	Value 1	100
Item 2	Value 2	200
Item 3	Value 3	300

5 Figures

To include a figure, use the following template (uncomment when you have an image):

6 Code Listings

Here's an example of a Python code listing:

Listing 1: Hello World in Python

```
1 def hello_world():
2     """A simple hello world function."""
3     print("Hello, World!")
4
5 if __name__ == "__main__":
6     hello_world()
```

7 Cross-References

You can reference equations, tables, and figures:

- Equation 1 shows the Gaussian integral
- Equation 2 is Euler’s identity
- Table 1 demonstrates table formatting

8 Conclusion

This template demonstrates the basic features of L^AT_EX. Happy typesetting!

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9 Citations and Bibliography

Here are some example citations from the bibliography:

- The foundational work on T_EX by Knuth [?]
- The L^AT_EX manual by Lamport [?]
- Einstein’s famous paper [?]
- The official L^AT_EX Project website [?]
- A sample conference paper [?]